

CPReady – Proposed Scope of Hardware Consultation with Sir Reyn

For initial hardware/electrical planning and preparation for physical fabrication

Purpose of this document: This document organizes the specific assistance we would like to request from Sir Reyn, an ECE graduate whom we are consulting for the initial electrical and hardware planning of CPreReady. The scope is intentionally focused on three areas: (1) initial hardware design, (2) circuit planning, and (3) circuit schematics.

Important note on the current design direction: The latest technical review/consultation by Sir Kherbin Cillooye Buenaventura is being strongly considered by the team and currently serves as our working direction. However, these proposed changes are not yet treated as irreversible final decisions. We will still discuss and verify them as a team and with the appropriate technical people, including Sir Reyn where the issue concerns hardware/electrical design.

Current Working Hardware Direction

Based on the latest technical review, the current proposed sensing configuration changes from the earlier FSR + MPU6050 setup to a dual-MPU6050 configuration: one MPU6050 on the chest/sensing side and a second MPU6050 as a base reference. The FSR is currently proposed to be removed. The rest of the physical structure generally remains based on the existing CPreReady concept.

- Outer compression pad/cover – polyester pad material.
- Force-distribution layer – thin EVA foam; current review suggests beginning bench testing at about 3 mm and adjusting experimentally.
- Chest-side sensing – MPU6050 #1.
- Base-side sensing/reference – MPU6050 #2.
- Rigid mounting plate and rigid structural base.
- Side electronics enclosures – hard protective case; 3D-printed PETG/ABS is the current proposed fabrication method, with a commercial plastic project box as fallback.
- Power – protected 18650 Li-ion cell with TP4056 charging module.
- Harness – polyester elastic webbing with adjustable buckle; top-and-bottom routing to be tried first, with X-harness as a possible modification if sensor drift occurs.

These are working directions from the latest technical review, not final instructions to fabricate. The purpose of Sir Reyn's consultation is to verify whether the proposed electrical and physical arrangement is technically appropriate before fabrication.

1. INITIAL HARDWARE DESIGN

Goal: Establish a buildable physical design before the hardware is fabricated.

A. Overall dimensions and proportions

- Determine the overall length, width, and thickness of the device.
-  Determine the dimensions of the central compression/sensing area.

- Determine the dimensions and available space for the side electronics enclosures.
- Determine the required clearance for components, wiring, connectors, battery, and charging access.
- Check whether the proportions provide adequate stability during repeated CPR compressions.

B. Layered structure

- Confirm the proposed layer sequence: outer pad → force-distribution layer → MPU6050 #1 → mounting/support structure → rigid base/manikin.
- Determine where MPU6050 #2 should be positioned as the base reference.
- Determine appropriate thicknesses and spacing for the layers.
- Verify that the physical structure will not interfere with sensor movement or introduce unnecessary displacement.

C. Component placement

- Specify the exact location and orientation of both MPU6050 boards.
- Specify the location of the ESP32, battery, charging module, and final board/perfboard or PCB.
- Determine which components should be placed inside the side enclosures.
- Plan the wiring routes so that wires are protected and do not interfere with the compression area.

D. Materials and fabrication specifications

- Finalize or recommend suitable materials for the pad, force-distribution layer, structural support, enclosure, harness, and mounting parts.
- Prepare a material/component list with quantities and basic specifications.
- Identify which parts can be commercially purchased and which parts should be fabricated or 3D-printed.
- Specify any material or fabrication that we should follow.

E. Mounting and maintenance

- Determine how the MPU6050s and other components will be physically secured.
- Determine how the side enclosures will be attached and opened for troubleshooting.
- Determine how the battery and charging module will be secured.
- Plan access points for charging, inspection, replacement, and maintenance.

Expected output from this part: a dimensioned physical design, layer/cross-section layout, component placement plan, material/component list, mounting plan, and fabrication specifications that the group can use as the basis for construction.

2. CIRCUIT PLANNING

Goal: Decide how the electrical components should interact before creating the final schematic or soldering the final prototype.

A. Overall electrical architecture

- Map the complete pathway from the two MPU6050 sensors to the ESP32 and then to the Bluetooth/mobile application.
- Determine which processing should occur on the ESP32 and what information should be transmitted to the application.

- Confirm that the proposed components can operate together within the intended system.

B. Sensor-to-ESP32 connections

- Determine the exact interface and wiring for MPU6050 #1 and MPU6050 #2.
- Resolve the I²C address configuration for the two MPU6050 boards so they can operate on the same bus without address collision.
- Determine the required SDA, SCL, VCC, GND, and AD0 connections.
- Confirm the correct voltage level and power requirements for the MPU6050 boards.

C. Power and charging system

- Determine the complete power pathway from the battery to the charging module and the electronic components.
- Confirm the appropriate protected 18650 + TP4056 configuration.
- Determine whether voltage regulation or other supporting power components are required.
- Check current/voltage compatibility, power distribution, and basic electrical safety.
- Determine the placement and wiring of the power switch/charging access if included.

D. Supporting components

- Identify any resistors, capacitors, pull-up resistors, protection components, connectors, or other supporting components that are actually necessary.
- Determine whether any component currently listed can be removed without affecting required functions.
- Specify the required wire type/gauge and connector type for the final prototype.

E. Prototype-to-final transition

- Determine what should be tested first on the breadboard.
- Determine which connections should later be transferred to perfboard or a custom PCB.
- Identify which parts of the circuit should remain modular for easier troubleshooting.

Expected output from this part: a complete circuit architecture/block diagram, power-distribution plan, sensor-to-ESP32 connection plan, pin assignment plan, list of required supporting components, and guidance for the prototype-to-final circuit transition.

3. CIRCUIT SCHEMATIC

Goal: Convert the approved circuit plan into a detailed electrical diagram that can be followed during assembly.

- Create/review the complete schematic of the CPReady hardware.
- Show the ESP32, both MPU6050 boards, battery, TP4056 charging module, power connections, ground connections, and all required supporting components.
- Indicate exact ESP32 pin assignments and the I²C connections for the two MPU6050 boards.
- Show the selected I²C address configuration for the two MPU6050s.
- Show component values where applicable, including resistors or other supporting components.
- Show connectors, switches, charging connections, and any protection/regulation circuitry.
- Provide clear labels for power rails, ground, SDA, SCL, and other relevant connections.

- If appropriate, provide a basic perfboard/PCB placement or wiring guide so the schematic can be translated into the physical assembly.
- Identify any electrical points that should be tested before final soldering.

Expected output from this part: a complete and labeled schematic, pinout/connection table, component-value list, and sufficient wiring/assembly guidance for the hardware builder.

What We Are NOT Primarily Asking Sir Reyn to Do

- Physically fabricate the entire prototype.
- Do the routine cutting, enclosure fabrication, soldering, or mechanical assembly that will be handled by the group.
- Develop the ESP32 firmware or Flutter application, which is being handled on the software side.
- Automatically finalize every design decision without team discussion. His role is to provide technical guidance, identify problems, and help us establish a sound initial design.

Division of Assistance

Person	Primary Role	Main Contribution
Sir Reyn + hardware team of the group	Initial hardware/electrical planning	Initial hardware design, circuit planning, detailed schematic, technical review/verification
Hardware team of the group	Hardware fabrication and assembly	Physical fabrication, mounting, wiring, soldering, enclosure/structural assembly, modifications based on approved design
Sir Kherbin + software team of the group	Software/firmware side	ESP32 firmware, sensor-data processing, CPR calculations, BLE communication, and mobile application integration

Important: Current Status of the Proposed Design

The dual-MPU6050 configuration, removal of the FSR, and the associated firmware architecture are the latest working recommendations presented by Sir Kherbin. The team currently agrees that this is the direction we want to seriously consider, but we are not treating it as completely final until the group discusses it and the relevant technical aspects are verified.

Basis from Our Existing CPReady Materials

- The latest technical presentation identifies the proposed change from FSR to a second MPU6050 and describes the dual-IMU sensing and firmware architecture.
- The latest technical review specifies the proposed hardware stack, protected 18650 + TP4056 power arrangement, EVA foam starting point, and enclosure/harness directions.
- The existing hardware plan already identifies the need for a physical layer structure, hard protective side enclosures, appropriate wiring, and a transition from breadboard prototyping to solderable final hardware.